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EEG Data Acquisition System and Analysis of EEG Signals

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***Abstract*— The main objective of this manuscript is to perform an EEG Data Acquisition, Analysis and Present the EEG signals. It consists of innovative pre-processing method by utilizing amplifier with high impedance and high CMRR. It also consists of Analog filters to remove the noise and to prevent from Aliasing. Analysis is done through FFT algorithm using MATLAB. Findings of the frequency of the signals corresponding to peak magnitude are put forward as the Presentation of EEG signals. As the biomedical signals are having low frequency and low amplitude, the pre-processing, filtering, conversion are done with accuracy, requires adequate sampling rate, processing speed, gain, power consumption and size.**

Index Terms: Aliasing, EEG Data Acquisition, EEG, sampling rate, pre-processing.

I. INTRODUCTION

A wide range of applications with biomedical signal processing. It is a fast growing field. The existence of the brain electric and magnetic fields are due to the interface of neurons in the brain. Electro-chemical signals, in the form, it takes place. The electrical activity of the cerebral cortex has the most affect on EEG. This is caused by its surface position. The classification of various patterns of band limited signals, according to frequency and amplitude range into five varieties, specifically named as gamma (γ), beta (β), alpha (α), theta (θ), and delta (δ) in descending order of frequency range. The different states (subject) of the brain indicate various brain wave patterns. These brain waves give different status of the brain such as feelings, emotions, the mood or any mental position. According to the person's feeling, work and thought the brain waves change [1, 11, 12].

The brain waves can also mostly changed by internal and external stimuli. With proper placement, electrodes on the scalp, the brain waves were transduced and also sensed [2]. These brain waves are recorded by noninvasive and invasive electroencephalogram techniques. These are utilized to diagnosis the disorders such as Alzheimer's Parkinson disease, brain tumors, dementia, and epilepsy, mental disorders, and sleep pattern change etc [3,13 ,14].

The types of brain waves with frequency range as shown in table 1. EEG is utilized for inspection of epilepsy, brain tumors, brain damage and other brain injuries. It is also used for determination of level of consciousness, and establishes death of brain [15, 16].

In this work, the brain waves are used, collected from wireless Neuro-Handset. The pre-processing is done with precision amplifier. The Analog filters are utilized to eliminate the noise in the signals. The LPF is enabled to minimize aliasing affect. This EEG Data acquisition system is built with the respective circuit diagram. The brain wave signals are analyzed and presented in the article.

The remaining part of the paper is planned as follows. In section II, brief descriptions of existing methodologies are illustrated. Section III consists of proposed method. In section IV analysis of the extracted signals are done. Section V is about the results through simulation. In section VI the conclusion of this article, discussing the outcomes.

TABLE I. TYPES OF BRAIN WAVES WITH FREQUENCY RANGE.

Sl. no.	Types of Brain Waves	Frequency (Hertz)
1.	Delta (δ)	0.50- 4.00
2.	Theta (θ)	4.00-8.00
3.	Alpha (α)	8.00-13.00
4.	Beta (β)	13.00-22.00
5.	Gamma (γ)	22.00-30.00

The remaining part of the paper is planned as follows. In section II, brief descriptions of existing methodologies are illustrated. Section III consists of proposed method. In section IV analysis of the extracted signals are done. Section V is about the results through simulation. In section VI the conclusion of this article, discussing the outcomes.

II. EXISTING METHODOLOGIES

The 10/20 International electrode placement system was explained. The adjoining electrodes are placed at either 10% or 20% distances on the cranium, where numbers 10 and 20 refers to distances. For high quality data for interpretation the EEG having proper function and recording electrodes are vital [3].

With any interfering signal, as the complete rejection. For the processing of these signals, the circuits are utilized. It must satisfy the safety patient's safety. The information contained in the original signal must be preserved, from the procedure of acquisition [4].

The acquisition procedure should maintain the data contained in the original signal. Drakulic et al. worked on the Intelligent Data Acquisition Unit (IDAU) [5]. It is completely equipped with a host interface and a DSP board [6].

The dataset which were collected from six patients with 48 channels EEG equipment using 10/20 regular system at

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512 Hertz sampling rate [7].These data was loaded and processed through MATLAB workspace. Three categories with various music tracks, such as motivational, soft and relax music. The data had been processed through the algorithm, to transform into commands to organize the audio tracks [8].

III. The Proposed method

The Planned method is consisting of EEG data buffering, acquisition, communication and MATLAB procedure. The overall flow chart of the real time brain waves analysis system as shown in Fig. 1. The buffering system is used to compensate the delay in acquiring the data. The cost-effective user pleasant portable Neuro-headset, such as Muse, Xwave, and emotive etc are utilized with especially available in market. In real time from state (subject) scalp, the EEG data can be acquired [10, 17].

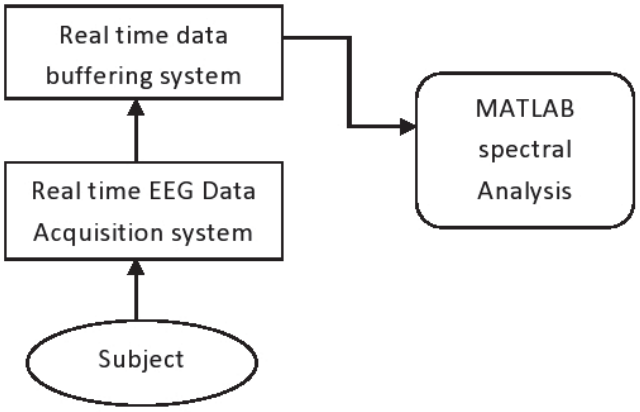


Fig. 1. Flow chart of the proposed method.

The amplitude and frequency vary quickly in biomedical signals. So the requirements of data acquisition system are high speed, flexible and powerful. The proposed method has a specialized amplifier for data acquisition .The Mosfet amplifier is utilized to amplify the signal. As the circuit may operate at low voltage (Approx. 1.5V, $V_{DD} \leq 5V$) and gives an accurate output. The range of the input signal may be from 10μV to 5mV. The output signals are recorded, through the computer system, or stored in the memory. These signals are directly being interfaced to MATLAB (R2016a) (9.0.0.341360) with SP tool box 7.2. As shown in figure 2.

In real time, this is done. With use of a dynamically-linked library (dII) routine. Actually, it requires some duration for acquiring data and computing it, in real time. By buffering software such as FieldTrip, and Datasuite, may be used to rectify this. Due to problems while using real time data, the sampled EEG data which are similar to the output of the real time system collected from PhysioNet and DEAP databases. These data can be interfaced with MATLAB, for analysis [9, 18].

The major diagnostic range of EEG signal is from 15 to 100mV. The frequency range is from 0.5 to 60 Hertz. The raw signals are captured by electrodes or a sensor. The signal is passed through an A1 amplifier. The Mosfet amplifier is used so that it requires low voltage. The signal range is from 10μV to 5mV. Amplification should be, done as close to the signal source. To remove aliasing affect, a Low pass filter is used. It is an II nd order filter with operating frequency of 35 Hertz. The signal is processed through a High pass filter with operating frequency of 0.5 Hertz. Again the signal is amplified by using second amplifier A2.When there are more signals, a Analog multiplexer is utilized [20].The signal is passed through the sample and hold circuit, where the signal is processed. Finally the signals are converted to digital, by using a A/D converter. The digital output may be given to data recorder etc.

IV. Analysis

Analysis is done in following steps. The single channel EEG data is collected from portable Neuro-Headset. These are such as Muse, Xwave and Emotive etc. The data is acquired using 14 electrodes for better bit rate and high resolution. The preprocessing is done to remove artifacts. It is passed through different filters. In real time, the processed data is interfaced by MATLAB. Initiating a dynamically-linked library routine. As BCILAB, BCI2000, and EEGLAB were efficient tools, are utilized for BCI Applications. If it is not possible, to acquire the sample EEG data, form real time data acquisition system. The taken samples must be similar to the real time systems. The data was loaded for analysis to MATLAB. By using FFT algorithm, the spectral analysis was done. By applying Load command, logical and looping commands, the peak amplitude and their respective frequency were found out. Among the signals brain wave band was foremost [19-22].Figure 3 indicate the flow chart of the proposed algorithm.

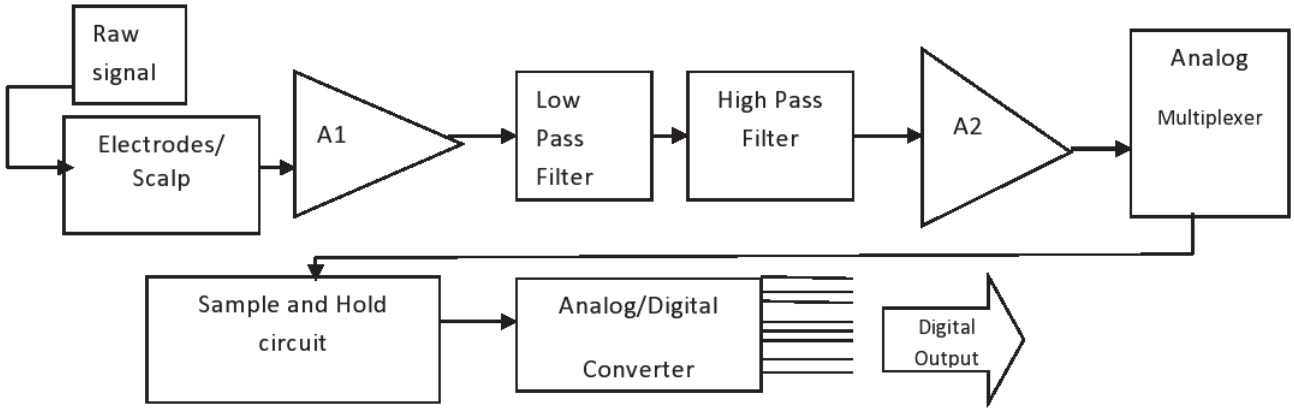


Fig. 2. EEG Data acquisition system

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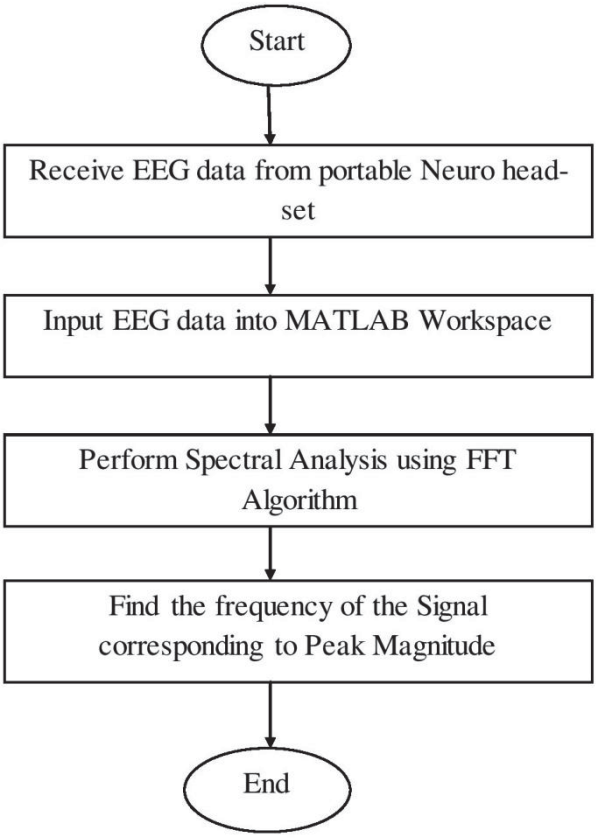


Fig. 3. Flow chart of the proposed Algorithm

V. RESULTS

The raw EEG signal and various waves' spectral resolution are illustrated. The data collected, produced raw EEG epochs. Relative PSD's were determined. The Fast Fourier Transform (FFT) algorithm was used. The frequencies bands are categorized into 5 classes. As illustrated in figure 4. The Gamma (γ) waves are connected with restless mental states. These are the highest frequency brain waves. The frequency of this wave is above 35 Hertz. The Beta (β) waves are related to mental status of focused, tensed and anxiety etc. These waves are having frequency range of 12 to 35 Hertz. The amplitude of these waves is nearly 20μV peak to peak .

Alpha (α) waves are associated with most stable state, and the frequency band is about 8 to 12 Hertz. The amplitude of this wave is around 10μV peak to peak. Theta (θ) waves are linked with deeply relaxed, having frequency range of 12 to 35 Hertz. Compared with delta waves, these are having low amplitude, but high frequency. The Delta (δ) waves have the frequency range nearly 0.5 to 4 Hertz. These signals are having high amplitude and low frequency, compared to theta waves. Infants having deep sleep are observed with these types of signals. The relative spectral densities of acquired EEG signals are as shown in figure 5 & 6.

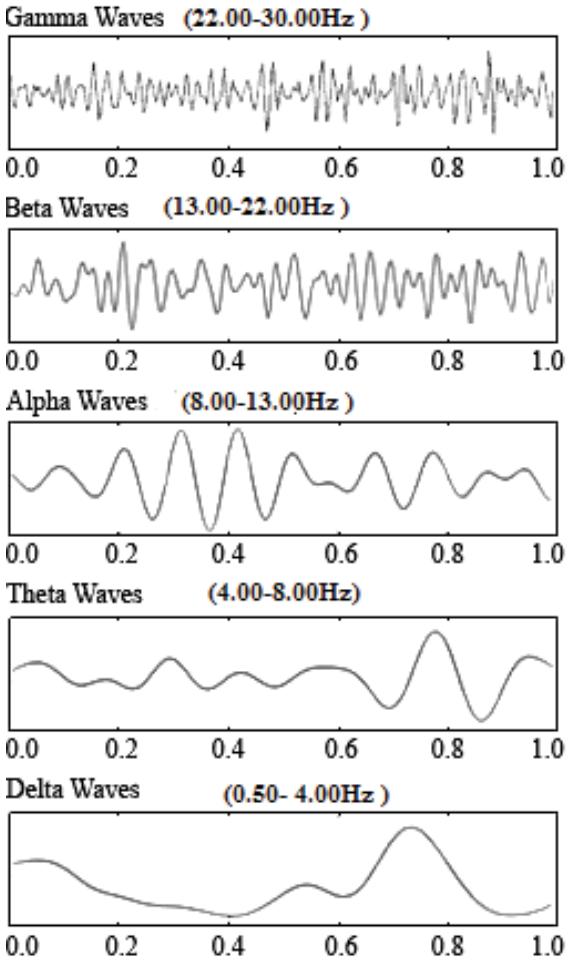


Fig. 4. Brain waves of a normal person.

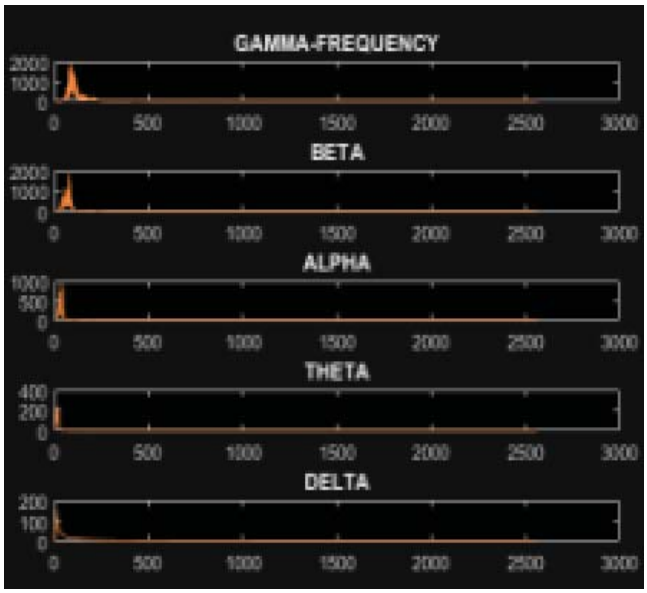


Fig. 5. Relative waveforms and Frequencies of acquired EEG signals.

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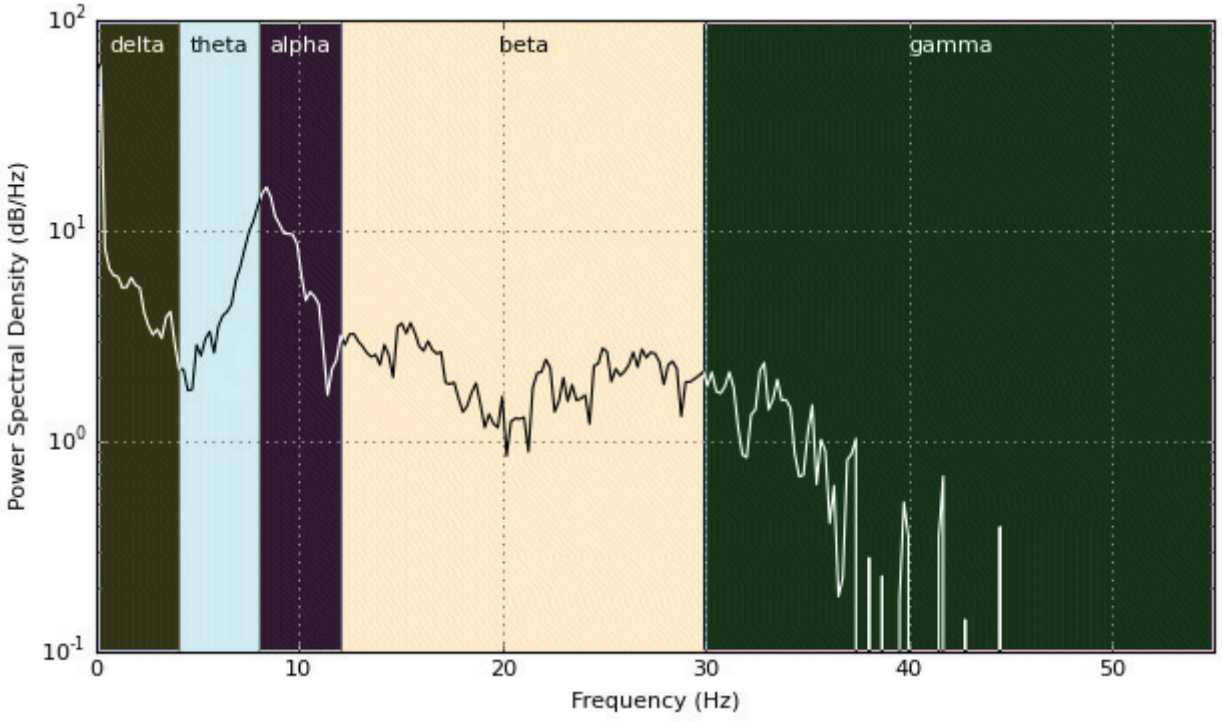


Fig. 6. PSD's of combination of brain waves.

VI. CONCLUSION AND FUTURE CONSIDERATIONS

In this article, the raw data has been preprocessed. EEG data acquisition system has been developed. Analysis of signals related to brain waves has been done effectively. The characteristics of various bands have been found. This approach can be extended for real time data acquisition system and development of more realistic nature of signals. The signals were collected with fewer channels, due to which occupied a small area of the brain. To control external devices, machines and to communicate. Various Brain Computer Interface (BCI) systems, home appliances, mobile robots, wheel chair etc are EEG-based brain controlled equipment. Mainly to assist, the disabled people. To advance biomedical applications, signal processing theory need to be used. The interface between the rehabilitation engineering and the biomedical signal acquisition systems is required. The human-machine interface is one of the prominent areas of research.

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